

Optiona Task

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Domain : AI & ML

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```
[3]
0s
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[34]
0s
df = pd.read_csv("transfusion.csv")
df.head()
```

▼

	Recency (months)	Frequency (times)	Monetary (c.c. blood)	Time (months)	whether he/she donated blood in March 2007
0	2	50	12500	98	1
1	0	13	3250	28	1
2	1	16	4000	35	1
3	2	20	5000	45	1
4	1	24	6000	77	0

```
[36]
0s
▶ print(df.isnull().sum())
```

▼

```
... Recency (months)      0
     Frequency (times)    0
     Monetary (c.c. blood) 0
     Time (months)        0
     whether he/she donated blood in March 2007 0
     dtype: int64
```

```

▶ print(df.shape)
print(df.info())
print(df.describe())

```

```

... (748, 5)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 748 entries, 0 to 747
Data columns (total 5 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Recency (months)                         748 non-null    int64
1   Frequency (times)                        748 non-null    int64
2   Monetary (c.c. blood)                   748 non-null    int64
3   Time (months)                           748 non-null    int64
4   whether he/she donated blood in March 2007 748 non-null    int64
dtypes: int64(5)
memory usage: 29.3 KB
None

```

	Recency (months)	Frequency (times)	Monetary (c.c. blood) \
count	748.000000	748.000000	748.000000
mean	9.506684	5.514706	1378.676471
std	8.095396	5.839307	1459.826781
min	0.000000	1.000000	250.000000
25%	2.750000	2.000000	500.000000
50%	7.000000	4.000000	1000.000000
75%	14.000000	7.000000	1750.000000
max	74.000000	50.000000	12500.000000

	Time (months)	whether he/she donated blood in March 2007
count	748.000000	748.000000
mean	34.282086	0.237968
std	24.376714	0.426124
min	0.000000	0.000000

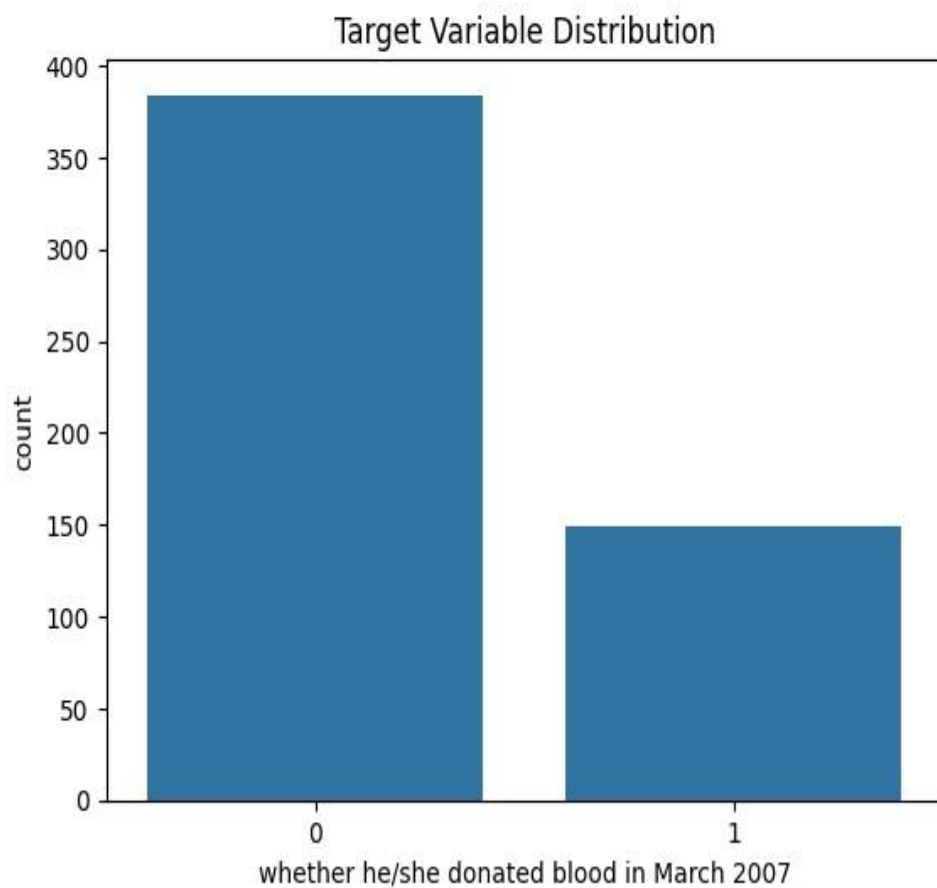
```
[7]  
Os ▶ print("Duplicate Rows:", df.duplicated().sum())  
  
df = df.drop_duplicates()
```

```
... Duplicate Rows: 215
```

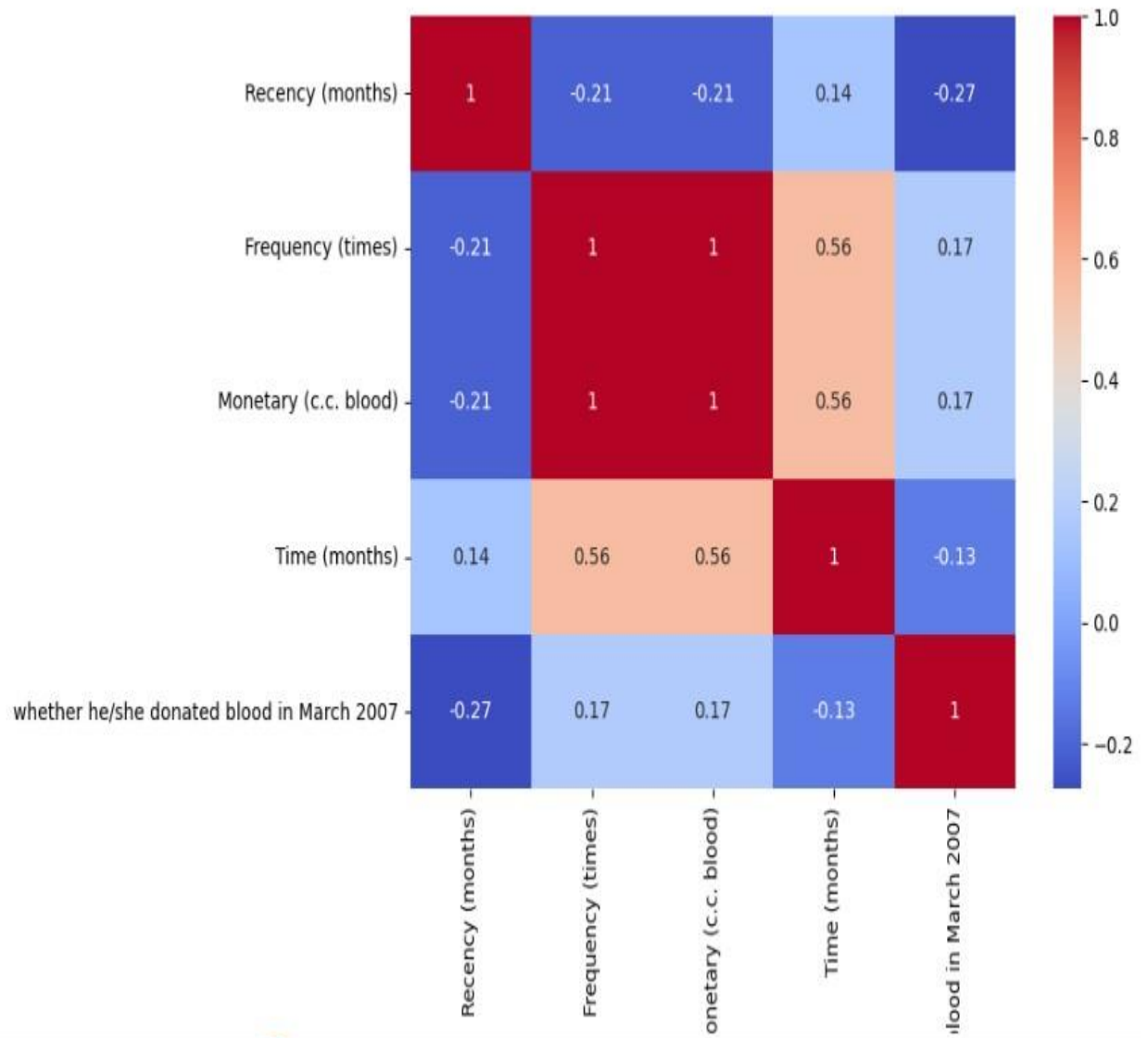
```
[38]  
✓ Os ▶ sns.countplot(x='whether he/she donated blood in March 2007', data=df)  
plt.title("Target Variable Distribution")  
plt.show()
```

▼

...

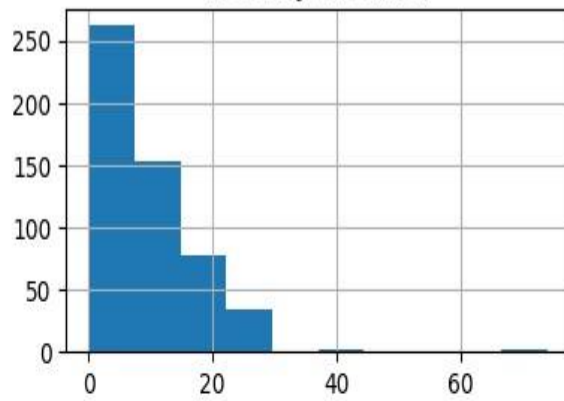


...

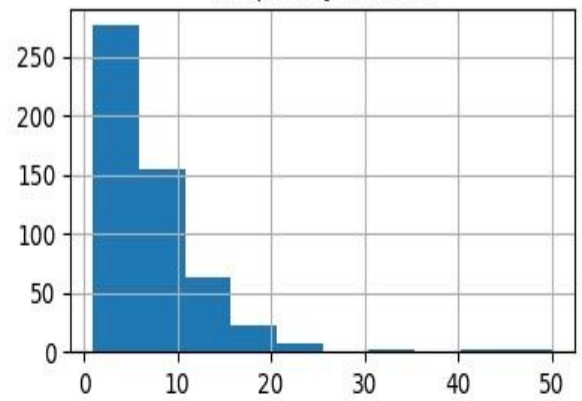


...

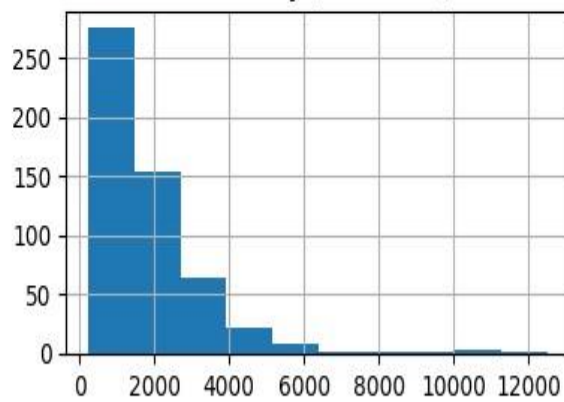
Recency (months)



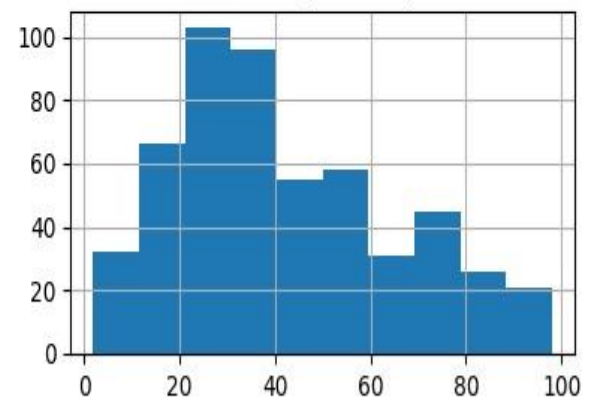
Frequency (times)



Monetary (c.c. blood)



Time (months)



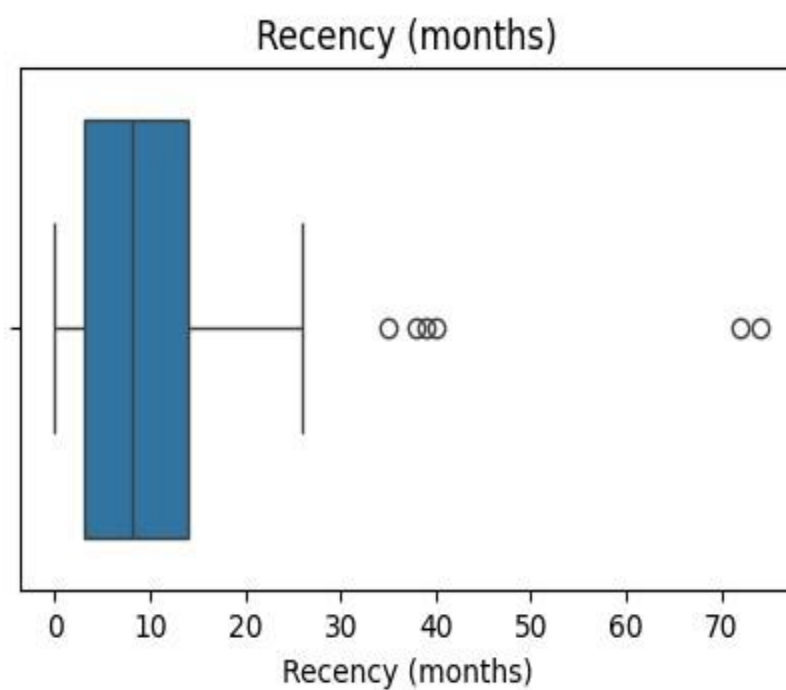
whether he/she donated blood in March 2007



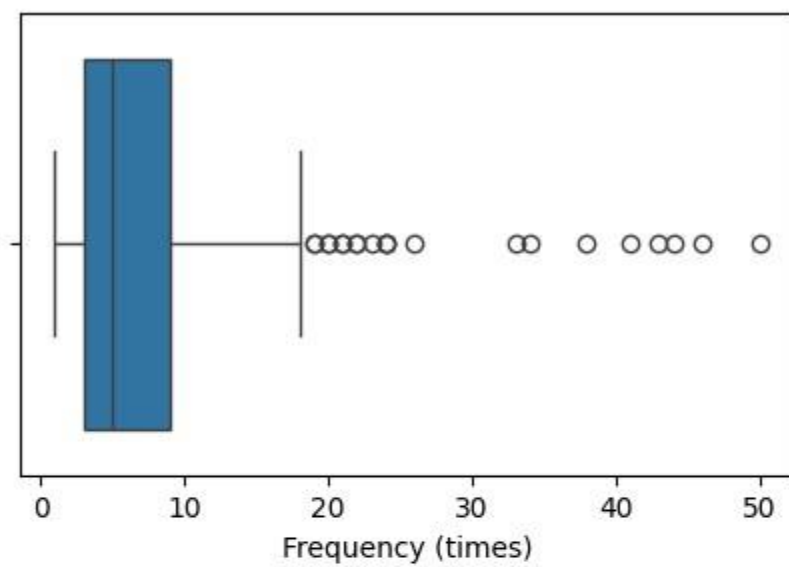
5

```
▶ for col in df.columns[:-1]:  
    plt.figure(figsize=(5,3))  
    sns.boxplot(x=df[col])  
    plt.title(col)  
    plt.show()
```

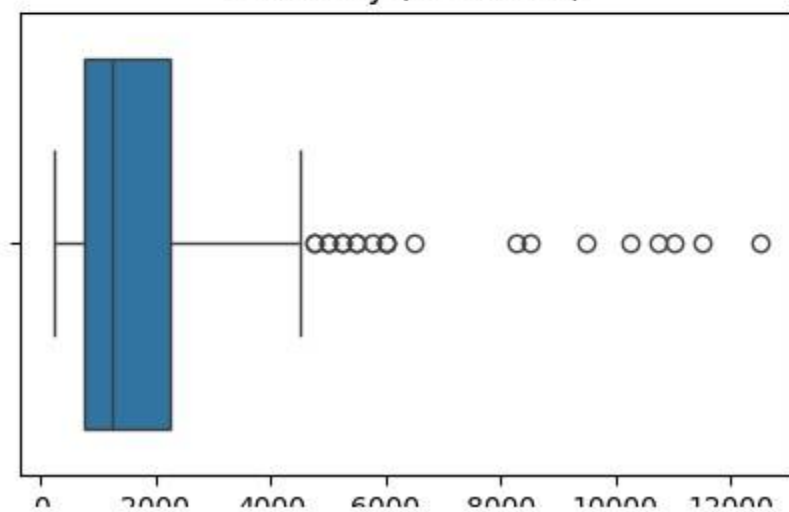
...



...



Monetary (c.c. blood)



[42]

✓ Os

```
X = df.drop("whether he/she donated blood in March 2007", axis=1)

y = df["whether he/she donated blood in March 2007"]
```

[43]

✓ Os

```
▶ from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

[44]

✓ Os

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

[45] `from sklearn.linear_model import LogisticRegression`

✓ 0s

```
model = LogisticRegression()
```

```
model.fit(X_train, y_train)
```

▼

▼ LogisticRegression ⓘ ?
LogisticRegression()

[46]

✓ 0s

```
y_pred = model.predict(X_test)
```

```
print(y_pred)
```

▼

```
[1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0  
 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0 0 0 1 0 0 1 0 1 0 0 0 0 0 0 0 0 0 1 0]
```

[47]

✓ 0s

▶ `from sklearn.metrics import accuracy_score`

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

▼

... Accuracy: 0.7663551401869159

[48]

✓ 0s



```
cm = confusion_matrix(y_test, y_pred)
```

```
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
```

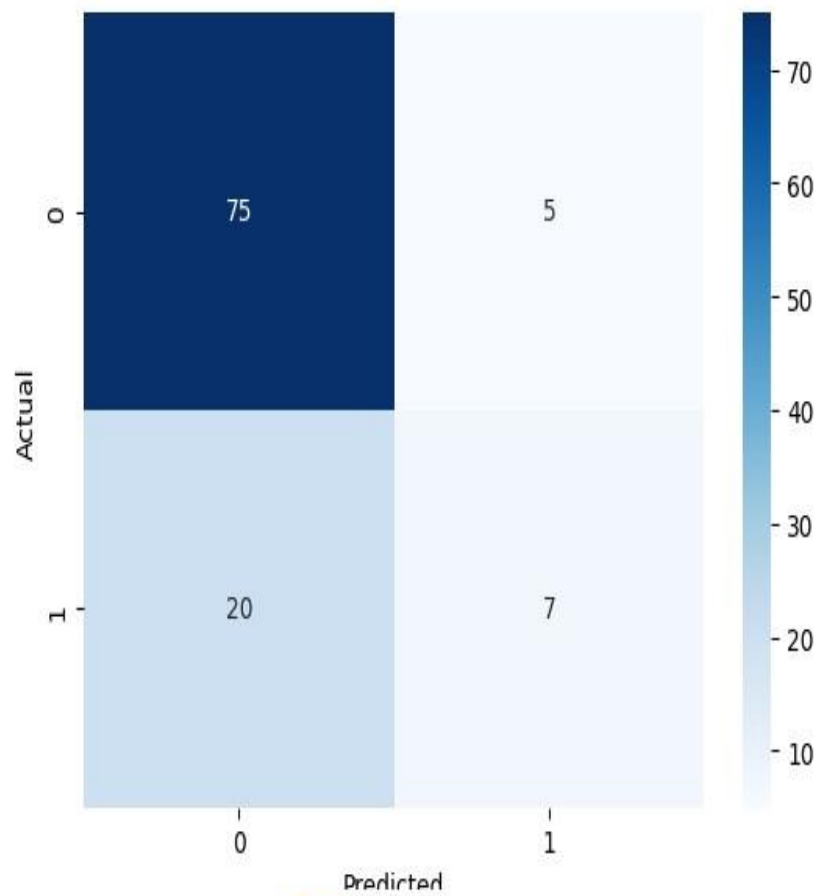
```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.show()
```

✓

...



[49]

✓ Os

```
import joblib

joblib.dump(model, "blood_model.pkl")
joblib.dump(scaler, "scaler.pkl")
```

▼

... ['scaler.pkl']

[50]

✓ Os

```
import streamlit as st
import joblib
import numpy as np

model = joblib.load("blood_model.pkl")
scaler = joblib.load("scaler.pkl")

st.title("Blood Donation Prediction")

recency = st.number_input("Recency (months)", 0)
frequency = st.number_input("Frequency (times)", 0)
monetary = st.number_input("Monetary (c.c. blood)", 0)
time = st.number_input("Time (months)", 0)

if st.button("Predict"):

    data = np.array([[recency, frequency, monetary, time]])

    data = scaler.transform(data)

    prediction = model.predict(data)

    if prediction[0] == 1:
        st.success("The person is likely to donate blood.")
    else:
        st.error("The person is not likely to donate blood.")
```

command:

```
*** streamlit run /usr/local/lib/python3.12/dist-packages/colab_kernel_launcher.py [ARGUMENTS]
2026-07-24 13:58:30.967 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.970 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.973 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.976 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.978 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.982 Session state does not function when running a script without `streamlit run`
2026-07-24 13:58:30.985 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.988 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.989 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.992 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:30.997 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.000 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.002 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.007 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.008 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.010 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.012 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.019 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.022 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.023 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.024 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.027 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.030 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.031 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.032 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.035 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.036 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.037 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
2026-07-24 13:58:31.039 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running
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```